

CURBRISK

Find the puddle before it floods the basement.

Street-level drainage risk scoring from LiDAR + pavement + 311.

Cities and insurers price flood risk with the wrong map.

FOR INSURERS

FEMA zones stop at the parcel.

Two homes on the same block — one at a low point with a failing catch basin, one on a crest — pay the same premium.

Underwriters have no measured, coordinate-level signal for which addresses pond. Cat modelers (RMS, Verisk) lack sub-zone drainage granularity.

FOR CITIES

Budgets follow complaints, not water.

Streets get repaved over unresolved drainage and fail again in 18 months. The loudest neighborhood wins the budget; the most hydraulically stressed one doesn't.

Climate change compresses storms into shorter bursts — 1970s-era basins now see 2020s-era loads.

CurbRisk: a measured, address-level flood score.

LiDAR

53.5M points
road-surface DEM

Hydrology

low-point detection,
catchment + 2-in ponding

Score

topo 40 / PCI 30 /
basin 20 / 311 10

Deliver

APS cross-section
+ address API

WHAT COMES OUT

A 0–100 CurbRisk score per street segment, a browser-viewable 3D cross-section of every flagged low point, and an API an underwriter queries by address in < 10 s.

Demo footprint: 1 Somerville tile • 5,080 pavement sections • 381 catch basins • 145 flooding 311s

Four data layers nobody else has integrated.

Measured terrain, not assumed

Mobile LiDAR captures the 2-inch crown differential that 1-meter DEMs miss. We see where water actually goes.

Pavement + drainage in one model

Cyvl PCI scores tell us where the road is already weak; terrain tells us where water lands on it. The intersection is where potholes reappear.

311 as ground truth

144 Somerville flooding reports validate every modeled low point. If complaints cluster on our high-risk segments, the model is right.

Autodesk-ready, browser-viewable

Every flagged low point exports as an APS cross-section an underwriter opens in a browser — and a civil engineer opens in Civil 3D.

The moat: nobody else holds measured LiDAR + pavement condition + drainage hydrology + Autodesk delivery in one stack.

Two markets, one engine.

PRIMARY • INSURANCE & RISK

Address-level flood pricing

- Underwriters: per-query API at point of sale
- Actuaries: portfolio loss exposure below the FEMA zone
- Cat modelers (RMS, Verisk): bulk dataset license

Differentiates premiums between the crest house and the low-point house on the same block.

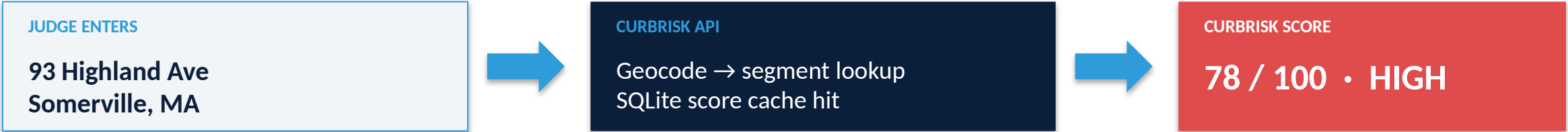
SECONDARY • MUNICIPAL DPW

Defensible capital prioritization

- Stop repaving over unresolved drainage
- Auto-generate MS4 / NPDES gap analysis
- Hand civil engineers a Civil 3D starting model
- Move to the front of FEMA BRIC / EPA CWSRF queues

Replaces complaint-driven politics with a ranked, auditable list.

Address in. Drainage risk out. Under 10 seconds.



RESPONSE PAYLOAD	
• Score breakdown	topo 32 • pavement 21 • basin 18 • complaints 7
• LiDAR geometry	low-point depth, 2-in ponding volume, catchment polygon
• 311 history	3 flooding reports within 100 m since 2023
• APS cross-section	browser-viewable 3D profile — no CAD software needed
• Recommended action	clear upstream basin; do not resurface until drainage resolved

WHAT'S NEXT

From one Somerville tile to a city-wide risk layer.

NOW

Hackathon MVP

One Somerville tile end-to-end. Address → score + APS cross-section in < 10 s.

NEXT

Paid pilot

\$20–50K fixed-scope study with a Somerville-style city. One paving package, one Autodesk model, one reference customer.

THEN

Insurer dataset

Bulk drainage-risk layer for one metro area, licensed to a cat modeler. Recurring revenue, no per-city sales cycle.

Stop pricing flood risk with a map that ends at the parcel.

Start pricing it with the one that ends at the curb.